

Tharun Rathod

B.Tech Electrical Engineering · IIT Roorkee · 4th Year (2023–2027)

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Profile

IIT Roorkee EE student who ships production systems solo – from a live edtech platform with 120+ users to a multi-backend compiler and MCP-based orchestration tools.

Education

Indian Institute of Technology, Roorkee

2023 – 2027

B.Tech, Electrical Engineering

Technical Skills

Languages	Python, TypeScript, JavaScript, C++, SQL
Backend	Node.js, Express, FastAPI, PostgreSQL, MongoDB, Prisma
Frontend	Next.js, React, Tailwind CSS
AI / ML	MCP, RAG Workflows, LLM Orchestration (Groq, Gemini, OpenRouter), Tool Calling
DSA & Systems	Algorithm Design, Graph Traversal (BFS/Dijkstra), Compiler Optimization, Docker, Git, REST APIs

Experience

Hiring Bazaar

Oct 2025 – Jan 2026

Web Developer Intern, IIT Roorkee Startup

Remote

- Built and shipped a live careers portal and end-to-end candidate application workflow; owned frontend QA and UI feedback loops across the onboarding flow.

R.K. IntelliServe

Dec 2025 – Feb 2026

Machine Learning Trainee

Gurugram, Haryana

- Engineered and automated DAG-orchestrated ML preprocessing and evaluation pipelines for production data-processing workflows.

Projects

Edcore – India's Academic Navigation Platform

2025 – Present

Solo Founder & Developer · Next.js, TypeScript, MongoDB, Vercel · edcore.tech

- Solo-built and shipped a live platform serving ~120 students, tracking 2,000+ colleges and 400+ exams across 15 academic fields with admission-rank predictions traced to official closing-rank data.
- Built end-to-end tutor/mentor booking flows with Resend-powered email confirmations and secure OTP-based student authentication, owning the full stack solo from concept to production.

Rune – Intent-Driven Algorithm Selection Compiler

2026

Python · Compiler Design · Parsing · Optimization · github.com/tharun99856/rune (★ 4)

- Built a DSL and optimizing compiler that infers problem constraints to auto-select optimal algorithms (BFS, Dijkstra, Kadane's, heaps, sorting), compiling to native C++; validated by 115 automated tests across 100+ problems.
- Deployed three interchangeable backends (Python interpreter, Numba LLVM-JIT, native C++ w/ -O2), verified output-identical; native C++ ran ~5× faster at 5M+ elements.

IITR Nexus – Unified Campus Intelligence Platform

2026

Next.js, TypeScript, Groq Llama 3.3 70B, MCP, Open Library API · iitrnexus.vercel.app

- Architected a multi-MCP router resolving composite user queries by identifying intent and dispatching to six parallel domain connectors, synthesizing results with source attribution.
- Deployed to production with real campus data – 18 academic departments, 80+ dated notices, live mess schedules across 15 hostels, and a multi-filter event catalog.

SmartAsset – Campus Resource Management Platform

2026

Next.js 16, TypeScript, Prisma, PostgreSQL, JWT Auth · smart-asset-rose.vercel.app

- Built a full-stack asset booking platform with conflict detection, admin approval workflows, and QR-based hand-over through a unified interface.
- Engineered atomic availability checking with database transactions and pessimistic locking, preventing double-allocation under concurrent admin approvals.

TravelChecker – Multi-Modal Travel Comparison Engine

2026

TypeScript, Node.js, MCP, Express, SQLite · travelchecker-resume.up.railway.app

- Built a live routing engine aggregating 5 transport modes across 117 Indian cities with weighted composite scoring, graceful multi-provider API fallback, and SQLite response caching.

Research & Publications

- The Capitulation Problem: Conditional Optimization in LLM Responses* – Sole Author, 2026. 484 trials across 5 domains; direct challenge reverses 64.6% of capitulations. [Preprint · arXiv]
- Specialist, Generalist, and Hybrid LLMs for Algorithmic Code Generation* – Sole Author, 2026. 75-problem benchmark, triaxial evaluation framework. IIT Roorkee.
- Dual-Transformer Cross-Attention Multimodal Emotion Recognition* – Co-Author, 2026. HuBERT + ViViT encoders, RAVDESS corpus. IIT Roorkee.
- Physics-Guided Diffusion Models for Synthetic Power Flow Data* – Co-Author, 2025. 91.4% feasibility vs. 41.2% baseline. IIT Roorkee.
- Phonological Fidelity in Indo-European Languages* – Sole Author, 2026. DOI: [10.5281/zenodo.19885744](https://doi.org/10.5281/zenodo.19885744).

Certifications & Achievements

MongoDB Core Concepts & Architecture (Jun 2026) · AIR 23,496 – JEE Advanced 2023 · Telangana EAPCET Rank 2614 · NCC Cadet, IIT Roorkee